



# COUNTERDIABATIC OPTIMIZED LOCAL DRIVING IN SPIN-SYSTEMS GROUND STATE PREPARATION

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## ABSTRACT

State preparation plays a pivotal role in numerous quantum algorithms, including quantum phase estimation. This project focuses on the optimization and benchmarking of a counterdiabatic driving protocol across three one-dimensional spin systems characterized by phase transitions, with particular focus on the axial next-nearest neighbor Ising (ANNNI) model. We introduce several enhancements, including the incorporation of Bayesian optimization and a comprehensive code base for computing various adiabatic gauge potentials. Our results demonstrate that this protocol consistently surpasses standard annealing schedules, often achieving performance improvements of several orders of magnitude. Notably, on the ANNNI model fidelities exceeding 0.5 are attainable in most cases. Furthermore, we observe that the optimized paths exhibit promising generalization capabilities to higher-dimensional systems, allowing for the extension of parameters from smaller models. This opens up possibilities for applying the protocol to higher-dimensional systems.



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## 1 INTRODUCTION

The efficient manipulation of quantum systems is a recurring problem in the realm of quantum technologies, with hardware implementations and simulations both being interested. Adiabatic Quantum Computing (AQC) and Quantum Annealing (QA) are two applications that heavily rely on time evolution protocols. AQC is based on adiabatic dynamics, where a system is driven such that the evolving state follows instantaneously the ground state of the system hamiltonian, naturally suppressing transitions towards excited states. QA protocols, instead, drop the assumption of adiabaticity and rely on quantum fluctuations to end up in the desired ground state.

However, limitations on the prepared state fidelity emerge in these frameworks. To be more specific, the literature on AQC provides a family of theorems that bound the minima evolution time required to achieve a final target fidelity in the preparation of a quantum state. In general, a slower preparation will increase the final state fidelity. From the hardware point of view, the requirement of using longer evolution times in AQC schedules is a serious limitation. In real near-term hardware the completion of protocols should be within the system's coherence time.

Furthermore, the instantaneous gap  $\Delta(t)$  between the ground state and the excited states is a crucial contribution in the adiabatic theorems, and it is required to be non-vanishing. Thus, schedules that imply a closing of the energy gap  $\Delta$  violate such hypothesis. As a result, the time evolution protocol is not robust against transitions towards excited states and the final fidelity drops consistently. For example, systems that undergo a first-order phase transition are known to produce an exponentially closing gap.

In the next section, we will introduce some techniques that aim to reduce losses occurring when the system undergoes fast deformations, far from the adiabatic limit. One particularly successful technique in this context is given by Counterdiabatic Driving (CD), which achieves the result by analytical compensation. While being an exact method, CD requires the solution of the full Schrödinger equation, which is not known in general. However, this condition can be relaxed by using an ansatz, which mitigates the diabatic losses, instead of canceling them completely. An interesting variant of CD protocols is the method introduced in [2], which is Counterdiabatic Optimized Local Driving (COLD), which will be the focus of this work. COLD combines CD with Quantum Optimal Control (QOC), aiming to reduce a loss function, which will be the prepared state infidelity.

After a discussion of all the relevant theoretical background, this contribution is meant to introduce a tool to handle the simulation of arbitrary spin system time-dependent dynamics and automatize the workflow required to apply COLD to such systems: the *colder* Python package [3].

## 2 METHODS

There exists classes of optimization problems whose solutions can be encoded in a quantum system's ground-state. This key intuition is used to set-up an experiment in which the system is driven towards its ground-state, eventually performing a readout to retrieve the solution. The protocols designed to solve such problems are mainly the Adiabatic State Preparation (ASP) and Quantum Annealing (QA). In this context, an optimal time-dependent control is a fundamental requirement.

Both protocols can be described within a time-evolution paradigm. Let  $H_0(t)$  be the time-dependent Hamiltonian acting on a quantum system. Suppose that the evolution starts at some initial time  $t = t_i$  in the ground-state  $|\psi_i\rangle$  of the Hamiltonian  $H_0(t_i)$ . Similarly, call  $|\psi_T\rangle$  the ground-state of the instantaneous Hamiltonian at a time  $t = t_f > t_i$ . The latter state is called “target state”, as it is the state we wish to obtain at the end of the time evolution. Let the total evolution time be  $\tau \equiv t_f - t_i$ ;





for convenience,  $t_0 = 0$ . Very often in literature, the time dependence is expressed as a function of a schedule parameter  $s \equiv t/\tau \in [0, 1]$ .

Even though the setup is quite general and allows to solve a variety of optimization problems, it is subject to limitations. Indeed, both protocols rely on a crucial assumption, which is that the system follows the instantaneous ground-state of  $H_0(t)$  during its evolution, eventually reaching the target state with null infidelity. The backbone of this assumption is a class of theorems known as “adiabatic theorems”, which characterize the constraints under which the assumption holds true.

In general, the main requirements are that the transformation is infinitesimally slow ( $\tau \rightarrow \infty$ ) and the instantaneous energy gap between the ground-state and the excited states is non-vanishing.

For instance, Refs [8, 4] report an instance of the adiabatic theorem that is specific for ASP. Calling  $|\phi(s)\rangle$  the actual evolving state at time  $s$ , if the instantaneous ground-state  $|\psi(s)\rangle$  of the Hamiltonian  $H(s)$  is separated by a gap  $\Delta(s) > 0$  from the excited spectrum (possibly degenerate, crossing permitted), then to guarantee that  $|\langle\psi(s)|\phi(s)\rangle| \geq 1 - \delta$ ,  $\forall s \in [0, 1]$ , it holds that

$$\tau \geq \frac{1}{\delta} \left( \int_0^s \left[ \frac{\|\partial_s^2 H(\sigma)\|}{\Delta^2(s)} + 7 \frac{\|\partial_s H(\sigma)\|^2}{\Delta^3(s)} \right] d\sigma + B \right) .$$

A straightforward consequence is that ASP (or QA) protocols require a long evolution time to achieve higher fidelities. Even so, in presence of a gap closure the assumptions do not hold anymore and the system evolves naturally towards excited states. For instance, a gap closure occurs spontaneously in systems which exhibit phase transitions.

In the context of state preparation and annealing, the most relevant accuracy metric is the final fidelity of the prepared state, which we wish to maximize. In the following sections, we will introduce two groups of methods that aim to achieve that goal through completely different approaches.

## A. QUANTUM OPTIMAL CONTROL

In Quantum Optimal Control, the dynamics of the system is manipulated to maximize or minimize an arbitrary metric. In the general time-evolution setup we have previously introduced, the system is prepared in the initial state  $|\psi_i\rangle$  and evolves in time towards the target state  $|\psi_T\rangle$ . A QOC problem is thus framed as the optimization of the Schrödinger equation

$$\dot{\psi} = f(t, \beta) ,$$

where  $\psi$  is the quantum wave function and  $\beta$  is the set of control parameters to be tuned.

The success metric is crucial for the optimization task, as it defines the optimization landscape to be explored. In practice, the metric is a cost function  $\mathcal{C}(\beta)$  that has to be minimized. As of common use in the context of state preparation, if  $|\psi_f(\beta)\rangle$  is the post-evolution state, the cost function is the infidelity of such state with respect to the target state, namely:

$$\mathcal{C}(\beta) \equiv 1 - |\langle\psi_T|\psi_f(\beta)\rangle|^2.$$

Rephrasing the dynamics in terms of Hamiltonians, the system undergoes an evolution controlled by the time-dependent Hamiltonian

$$H_{\text{QOC}}(t, \beta) = H_0(t) + f(t, \beta) \mathcal{O}_{\text{opt}} . \quad (1)$$

The first term, usually addressed as the *bare* Hamiltonian, describes the dynamics of the target system without regard for optimization. The second term is an additional driving term, which makes use of the operators  $\mathcal{O}_{\text{opt}}$  to provide additional degrees of freedom that are used by the optimizer subroutine to minimize the cost function.





## B. COUNTERDIABATIC DRIVING

Various protocol designs, which are called *shortcuts to adiabaticity*, have been proposed in order to mitigate the limitations prompted by the adiabatic theorems. The general target is to shorten as much as possible the evolution time and avoid gap closures in the spectrum. Among the shortcuts-to-adiabaticity methods, Counterdiabatic Driving (CD) is a promising candidate, which is formally able to overcome this problem through an ingenious choice of the additional driving term. The basic idea behind CD is to boost any adiabatic process by adding a CD Hamiltonian (the *adiabatic gauge potential*) that suppresses transitions between the system eigenstates:

$$H_{CD}(t) = H_0(t) + i\hbar \underbrace{\sum_n (|\partial_t n\rangle \langle n| - \langle n|\partial_t n\rangle |n\rangle \langle n|)}_{\mathcal{A}_{AGP}(t)},$$

with  $|n\rangle \equiv |n(t)\rangle$  being the  $n$ -th eigenstate of the instantaneous Hamiltonian  $H_0(t)$ . To construct the additional driving term  $\mathcal{A}_{AGP}$  it is necessary to have prior knowledge of all the eigenstates at all times during the system's dynamics. This represents a huge limitation from the computational and even from the experimental point of view.

To overcome this complexity, it is common use to approximate the adiabatic gauge potential  $\mathcal{A}_{AGP}$  using a suitable, local ansatz. The protocols designed on this idea belong to the class of Local Counterdiabatic Driving (LCD). The system dynamics is controlled by the time-dependent Hamiltonian

$$H_{CD}(t) = H_0(t) + \underbrace{\sum_j \alpha_j(t) \mathcal{O}_{LCD}^{(j)}}_{\mathcal{A}(t)} \quad (2)$$

$$s.t. \quad \mathcal{A}(t) \simeq \mathcal{A}_{AGP}(t), \quad (3)$$

where  $\{\mathcal{O}_{LCD}\}$  is a set of LCD operators. In this framework,  $\alpha(t)$  has to be optimized in order to approximate the adiabatic gauge potential (Eq.3).

## C. COUNTERDIABATIC OPTIMIZED LOCAL DRIVING

Counterdiabatic Optimized Local Driving (COLD) is a method [2] that combines the two approaches introduced in the two latter paragraphs. Consider the CD Hamiltonian of Eq. 2 and replace the bare Hamiltonian  $H_0$  with the QOC Hamiltonian  $H_{QOC}$  of Eq. 1. To apply this in the context of ASP and QA, the control function must vanish at boundary, i.e.,  $f(0, \beta) = f(\tau, \beta) = 0$ .

Explicitly, the Hamiltonian is written as

$$\mathcal{H}_{COLD}(t) = \underbrace{(H_0(t) + f(t, \beta) \mathcal{O}_{opt})}_{H_\beta(t)} + \alpha(t, \beta) \mathcal{O}_{LCD} \quad (4)$$

Notice that  $\alpha(t) \rightarrow \alpha(t, \beta)$  to remark that the optimization of the gauge approximation through  $\alpha(t)$  is now depending on the choice of the QOC parameters  $\beta$ , which are changed by the optimization routine when minimizing the cost function  $\mathcal{C}(\beta)$ .

The coefficient of the QOC term is the control function

$$f(t, \beta) = \sum_{k=1}^{N_k} \beta^k \sin(\pi k t / \tau) \quad ,$$

which represents a parameterized pulse. The optimization task consists in determining the coefficient  $\beta^k \in \beta$  of the  $k$ th frequency of the control function.





In ASP and QA problems, it is common to write the bare Hamiltonian as

$$H_0(t) = H_i + \lambda(t)(H_f - H_i)$$

where  $\lambda(t) : [0, \tau] \rightarrow [0, 1]$  is a monotonically increasing function (called *schedule function*),  $H_i \equiv H_0(0)$  and  $H_f \equiv H_0(\tau)$  trivially. It is common to use a linear schedule function  $\lambda(t) = t/\tau$ , but in this work the schedule function will be

$$\lambda(t) = \sin^2 \left( \frac{\pi}{2} \sin^2 \left( \frac{\pi t}{2\tau} \right) \right) \quad (5)$$

whose first  $\dot{\lambda}$  and second derivative  $\ddot{\lambda}$  vanish at the beginning and end of protocol.

Once that  $H_\beta$  is determined ( $\beta$  is fixed), the parameters  $\alpha$  are optimized to realize the condition of Eq. 3. For this reason one could say that the adiabatic gauge potential ansatz is path-dependent. The approximation of the adiabatic gauge potential is realized following the methods of Ref. [7]. Briefly, it consists of defining the quantity

$$G = \partial_t H_\beta + \frac{i}{\hbar} [\mathcal{A}, H_\beta] \quad (6)$$

which satisfies the closed-form equation  $[G, H_\beta] = 0$ . Eventually, the condition of Eq. 3 can be cast as the minimization of the Hilbert-Schmidt norm of  $G$ , or equivalently to the minimization of the action

$$S(\mathcal{A}) = \text{Tr} [G(\mathcal{A})^2] \quad (7)$$

This condition is equivalent to a set of  $\dim(\alpha)$  equations that have to be solved  $\forall t \in [0, \tau]$ .

## D. ENHANCEMENTS AND APPLICATION ON SPIN SYSTEMS

It is clear that the minimization of Eq. 7 has to be carried out specifically for each combination of system ( $H_0$ ), control terms ( $\mathcal{O}_{\text{opt}}$ ) and gauge potential ansatz ( $\mathcal{A}$ ). For instance, the original work [2] carries out the computations for a linear chain 1D Ising model. In general, this procedure is non-trivial, especially when facing more complex spin systems.

Our main contribution to the application of COLD consists in a symbolical solution framework of Eq. 7, which is implemented through an extension of the *SymPy* library [5] to efficiently compute the commutator of operators that involve Pauli spin matrices applied on individual sites. The resulting framework [3] is able to automatize the workflow of COLD for arbitrary systems and ansatz choice.

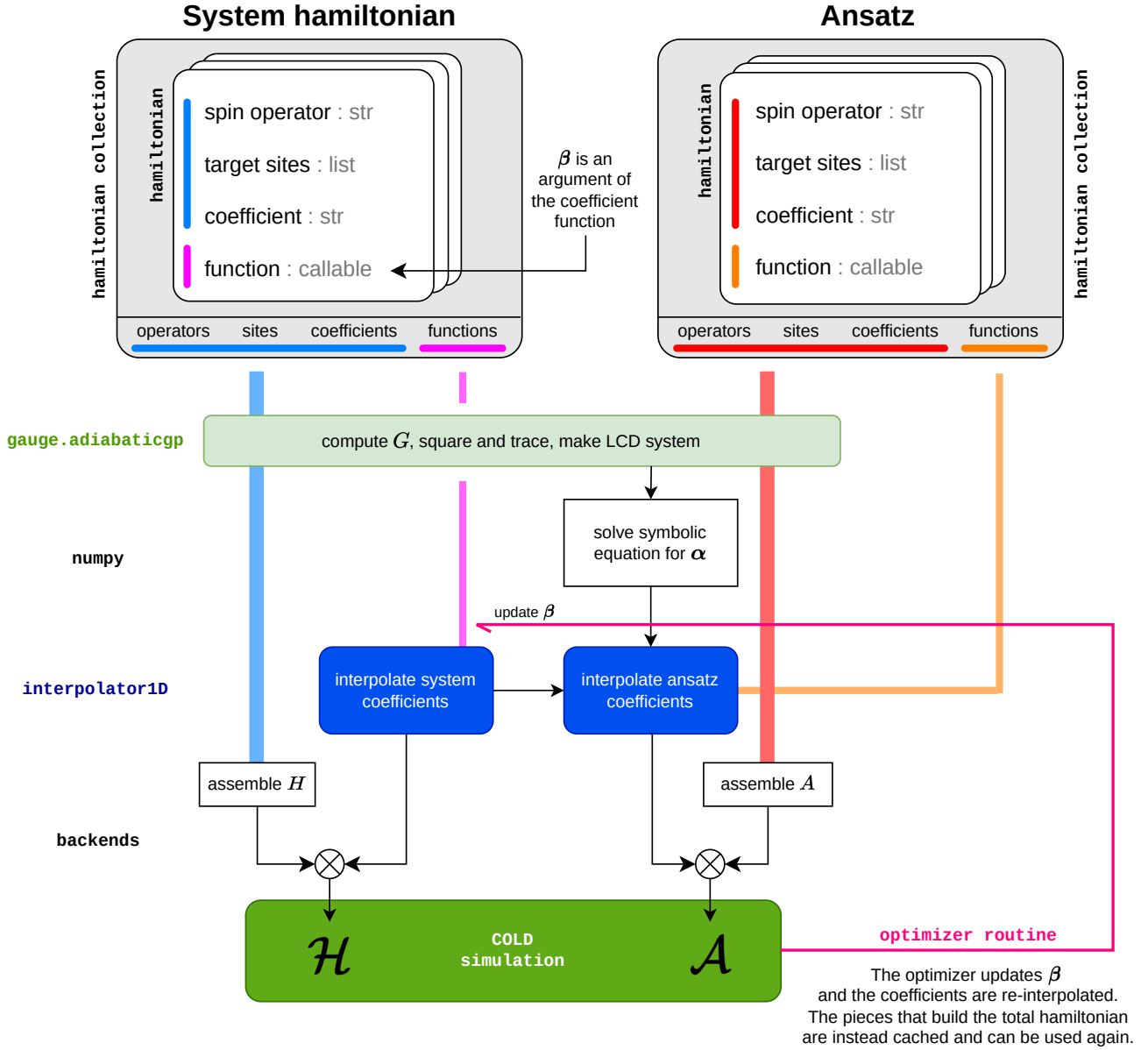
A further enhancement introduced in this work consists in the replacement of the loss optimizer. In the original work [2], Powell minimization [6] is used to explore the parameter space. In our framework, the Powell optimizer is replaced by a Bayesian optimizer [1] which samples the loss function by changing the parameter  $\beta$  in a pre-defined interval, and returns the optimal parameter. In our experience, the Bayesian optimizer does not reduce drastically the number of iterations that are needed to select an optimal parameter. Instead, an advantage in escaping from local minima is observed. In this regard, other optimization strategies might get stuck in local minima, and the only possible mitigation is to execute several times the optimization with independent initial parameters and select the best simulation.

The present contribution aims to introduce these tools as a Python package, *colder*[3], which handles the AGP computation symbolically and performs the optimization steps required by the COLD method. We will show, in the next section, a general picture of the framework components and how it handles the input of general spin systems.

## 3 THE COLDER PACKAGE

The aim of this project is to apply the COLD method to general quantum spin systems. As described in the previous sections, this machinery requires some optimization operations which are specific for





**Figure 1:** Schematics of the *colder* package inner workings.

any spin system and ansatz choices. An example could be the minimization of Eq. 7, which is to be carried out analytically. Furthermore, an optimizer has to be able to change the QOC parameters and call a new simulation. This requirement hints the necessity to integrate such operations in the framework. To make the workflow easier and as general as possible, a Python package is introduced.

*Colder* (COLD annealER) is a package [3] that handles all the necessary symbolical and numerical operations that are required to build an optimized schedule for ASP and QA problems. A schematics of its inner workings is presented in Figure 1, even though the details are not going to be discussed at such level in this document. Nevertheless, it is better to keep in mind a sketch of the main steps:

1. **The user prompts a system Hamiltonian  $H_\beta(t)$  and the ansatz choice  $\mathcal{A}(\alpha)$ .** From the point of view of the implementation, both objects are formally the same entity (i.e. an Hamiltonian acting on a spin system), thus they are handled using the same class (`hamiltonian_collection`). The distinction between the system and the ansatz is made explicit only in the next step of the workflow.





- The two Hamiltonians are prompted as a sum of terms. Each term is in general written in the form

$$Y(t) \cdot \sum_{i \in [\text{target sites list}]} \hat{\sigma}_i^{(\text{operator})}$$

The class `physics.hamiltonian` is introduced to input all the required entities: a coefficient symbol “ $Y$ ” and its functional dependence  $Y(t, \beta, \dots)$ , a list of target sites and the operator acting on them. The direct sum (+) of `hamiltonian` objects will create automatically a new `hamiltonian_collection` object.

- The coefficient has to be introduced as a *string* symbol and as a *callable* function. The string is used to identify the symbol in symbolic computations, whereas the function is to be called in the numerical simulation.
- The coefficient function is a *callable* function, whose first argument must be the time variable. Other arguments are allowed (and recommended), such as the optimizable parameters  $\beta$ , or the total evolution time  $\tau$ . It is important to remember that all the coefficient functions must share the same I/O format and can be called with vectorialized time argument.
- A list of target sites is required to apply the Pauli spin operators. For instance, say that the user wants to implement the Hamiltonian  $\sum_i^N \sigma_i^z$ . The list of target sites would be  $[0, 1, \dots, N-1]$  and the operator string ‘Z’. The allowed letters are ‘X’, ‘Y’ and ‘Z’.
- To implement  $k$ -body interactions, the operator string must be a string of  $k$  letters. In this case, the list of target sites must be prompted as a list of tuples, each element of the tuple being associated to the  $k$ th symbol of the string.

We provide two examples:

$$\begin{aligned} \text{list} = [(0, 1), (1, 2)], \text{ operator} = \text{'XY'} &\longrightarrow \sigma_0^x \sigma_1^y + \sigma_1^x \sigma_2^y \\ \text{list} = [(0, 1, 3), (1, 2, 3)], \text{ operator} = \text{'ZYZ'} &\longrightarrow \sigma_0^z \sigma_1^y \sigma_3^x + \sigma_1^z \sigma_2^y \sigma_3^x \end{aligned}$$

- More terms can share the same coefficient.

2. **The coefficients  $\alpha$  are symbolically determined through the minimization of Eq. 7** ( $\text{Tr}[G(\mathcal{A})^2]$ ). A custom class (`gauge.adiabaticgp`), which takes the system and ansatz Hamiltonians as inputs, is specialized for this task.
3. **The coefficients of the Hamiltonians are numerically interpolated in time.** Since the ansatz coefficient  $\alpha(t)$  depend on the system Hamiltonian coefficients, the coefficients of are interpolated at first. Secondly, the ansatz driving coefficients are interpolated from the symbolic solution provided in the previous step.  
The interpolation, even though is not strictly necessary, is a strongly suggested step in this workflow, since the symbolic solution can be numerically sensitive if the ansatz is not optimal. This numerical sensitivity might arise from near-singularities which can emerge in the optimization.
4. **The full Hamiltonians are built**, merely by multiplying each term by the right coefficients. The exact way this happens is depending on the *backend* chosen by the user. Backends on `scipy`, `cupy` and `qibo` are provided.

## A. USAGE EXAMPLES

### i. Hamiltonian Input

This section will be a brief overview of the package usage, with particular focus on the input of a spin system. A notebook is provided in the GitHub repository attached to this work [3].



The `physics` module of `colder` provides the interface to build generic systems of spins. Suppose that one would like to use the following Hamiltonian:

$$\mathcal{H}_z = A \cdot (\sigma_0^z + \sigma_1^z + \sigma_2^z)$$

Then, its realization will be

```
import colder.core.physics as cphys
H_z = cphys.hamiltonian('Z', [0,1,2], coeff='A')
```

In the same fashion, two body interactions are prompted:

$$\mathcal{H}_{int} = J \cdot (\sigma_0^x \sigma_1^x + \sigma_1^x \sigma_2^x) \quad \longleftrightarrow \quad \begin{aligned} \text{H\_int} &= \text{cphys.hamiltonian}(\text{'XX'}, [(0,1), (1,2)], \\ &\text{coeff}=\text{'J'}) \end{aligned}$$

The direct sum of `hamiltonian` objects will create a sum of terms that are collected in the `hamiltonian_collection` object:

$$\mathcal{H}_{sys} = \mathcal{H}_z + \mathcal{H}_{int} \quad \longleftrightarrow \quad \text{H\_sys} = \text{H\_z} + \text{H\_int}$$

Notice that in the previous code snippets no functional dependence of  $A$  and  $J$  has been specified, thus these objects cannot be used for a COLD simulation. Nevertheless, with this simple actions it is possible to build the Ising Hamiltonian that is going to be tested in the next section.

## ii. Example of implementation: the Ising model

To provide a working example of COLD simulation, let us replicate the Ising model experiment of the reference paper [2]:

$$\mathcal{H}_0^{(\text{Ising})} = X(t) \sum_i \sigma_i^x + J(t) \sum_i \sigma_i^z \sigma_{i+1}^z + Z_\beta(t) \sum_i \sigma_i^z \quad (8)$$

$$\text{with} \quad X(t) = 10 \cdot \lambda_\tau(t), \quad J(t) = -1, \quad Z_\beta(t) = 0.02 + f(t, \beta)$$

The following code snippet (Listing 1) is a complete example of system Hamiltonian implementation. Each term of the system Hamiltonian is created as a `hamiltonian` object and eventually summed to the others. The coefficient functions are regular Python callable functions and are linked to each term using the argument `coeff_function`.

```
import colder.core.physics as cphys
import colder.models.lattice as cll

# this is just a tool to get the correct spin indices
nspin : int = 5
lattice = cll.chain(nspin = nspin, periodic = False)

# lambda(t) as defined in the reference paper
def scale_f(t, tau):
    """Define the scaling function [0,tau] -> [0,1]."""
    return np.sin( (np.pi/2)*(np.sin(np.pi*t/(2*tau))**2) )**2

# f(t, beta)
def f_qoc(t : float, tau : float, beta : np.ndarray):
    """control function f(t, beta) to use for qoc"""
    k = np.arange(len(beta)) + 1
```



```

    return np.sum( beta*np.sin( np.pi*k*t/tau ), axis=1 )

# input of system Hamiltonian -----
def xf(t, tau, beta):
    return 10*scale_f(t, tau)

H_X = cphys.hamiltonian('X', lattice.single_site, coeff = 'X', coeff_function=xf)

def Jf(t, tau, beta):
    return -1*np.ones_like(t)
    #             ^^^ trick to be vector-safe
def zf(t, tau, beta):
    return 0.02*np.ones_like(t) + f_qoc(t, tau, beta)

H_J = cphys.hamiltonian('ZZ', lattice.nearest_neighbor, coeff = 'J', coeff_function=Jf)
H_Z = cphys.hamiltonian('Z', lattice.single_site, coeff = 'Z', coeff_function=zf)

H = H_J + H_Z + H_X  # the final system hamiltonian

```

**Listing 1:** Implementation of Ising model (Eq. 8) from [2].

The same procedure has to be repeated for the ansatz Hamiltonian. To simplify this document, the implementation of the ansatz is not reported, but can be obtained from the example notebook in the package repository.

Eventually, the COLD simulation can be initialized using the simulation package, which will handle all the necessary operations in background.

```

import colder.simulation.cold as symcold

# H as above
ansatz = ... # look at the notebook

tau : float = 0.01 # annealing time

csym = symcold.cold(
    system = H, ansatz = ansatz, annealing_time = tau, nspin = nspin,
    # the following parameters are to be passed at the system coefficient functions
    system_fargs = {'tau' : tau, 'beta' : np.zeros(1) }
)

```

**Listing 2:** The COLD simulation is initialized by prompting the system and ansatz Hamiltonians, the total annealing time, the number of spins and the argument of the control functions.

## 4 RESULTS

### A. ISING MODEL

To set a baseline for the applications of *colder* and to test its routines, the main results of the reference paper [2] have been replicated within our framework. In particular, we focus on the state preparation

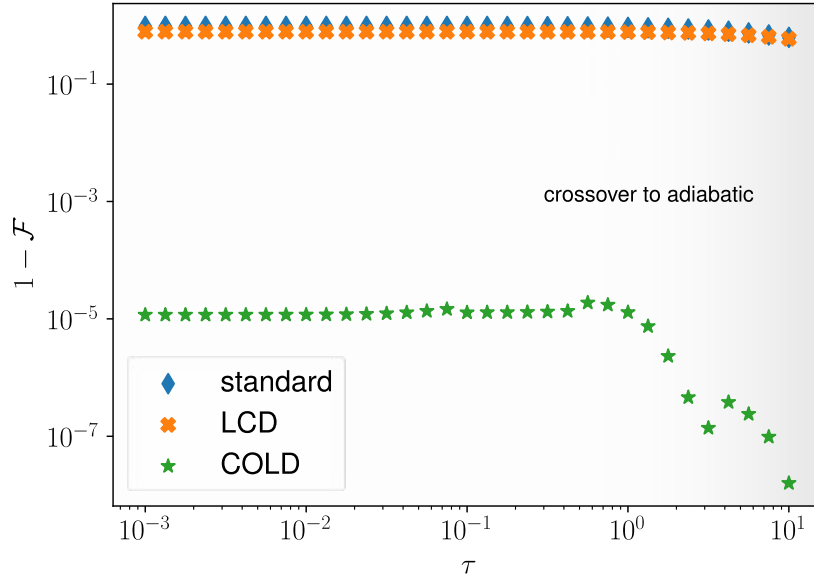




of a 1D Ising spin chain in the presence of a transverse and longitudinal field:

$$H_f^{(\text{Ising})} = -J^* \sum_j^{N-1} \sigma_j^z \sigma_{j+1}^z + Z^* \sum_j^N \sigma_j^z + \lambda(t) X_f \sum_j^N \sigma_j^x \quad (9)$$

The annealing protocol is set to prepare the ground state across the Ising phase transition, sweeping  $\lambda$  from 0 to 1.  $Z^* = 0.02$  is a small field meant to break the ground state degeneracy.  $X_f = 10$ . The total evolution time is  $\tau$ , which is changed in the range  $[10^{-3}, 10^1]$ .



**Figure 2:** Preparation infidelity for the unassisted protocol (labelled as ‘standard’), LCD alone and COLD.  $N = 5$ .

In Figure 2, the infidelity of the preparation for a schedule of total evolution time  $\tau$  is shown, comparing the results for a standard (unassisted), LCD only and COLD schedules. The data hints that COLD is able to prepare states with an infidelity reduced by orders of magnitude with respect to unassisted protocols and LCD. Eventually, increasing the values of  $\tau$  brings the systems towards an adiabatic regime in which the fidelity naturally improves through all the tested methods.

## B. BEYOND THE ORIGINAL WORK

In the following section, we discuss the application of COLD method for the ground-state preparation of a non-trivial spin system: the ANNNI model. In general, the relative improvement of the prepared state fidelity (which we will define as  $\mathcal{R}$ ) remarkably follows a pattern depending on the quantum phase of the prepared ground-state, with consistent improvement in non-ferromagnetic phases.

In a typical QA protocol, the spin system is prepared in the ground-state of  $H_i = -\sum_i^N \sigma_i^x$  and the evolution is carried out towards a target Hamiltonian  $H_f^{(\text{sys})}$ , which identifies the specific model (see Eq. 14 in this case). The final Hamiltonian will have two free parameters that are changed in order to prepare ground-states in the different phases. Eventually, the COLD optimization is carried out independently for each parameter combination.

The prepared state infidelity is minimized by tweaking one coefficient of the control function (thus  $N_k = 1$ ). The total evolution time is set to be  $\tau = 0.01$ . A chain of  $N = 5$  spins is considered, as a first baseline for all benchmarks, before eventually considering larger spin numbers.



As previously mentioned, the control operators  $\mathcal{O}_{opt}$  play a fundamental role, as well as the choice of a suitable ansatz  $\mathcal{A}$  for the gauge potential. Regarding the choice of the control operators, several preliminary tests have been performed. It has been observed that the ANNNI model achieves good fidelities with a simple local control operator  $\sigma^z$  on each spin.

The ansatz choice is crucial for the success of LCD and COLD. From a theoretical point of view, it is fundamental to assure that an ansatz does not produce a vanishing commutator with the  $H_\beta$  Hamiltonian in Eq. 6. Furthermore, the choice of the ansatz is also driven by experimental considerations. Indeed, local interactions are easier to implement and map into real hardware, and for this reason, it is preferable to keep the ansatz relatively simple. The most simple choice of the ansatz is a local field:

$$\mathcal{A}_{\text{local}} = \alpha_1 \sum_i \sigma_i^y . \quad (10)$$

Following the footsteps of the reference paper [2], we test a second order ansatz ('second order' meaning that it introduces two-body couplings):

$$\mathcal{A}_{\text{near}} = \alpha_1 \sum_i \sigma_i^y + \alpha_2 \sum_i (\sigma_i^x \sigma_{i+1}^y + \sigma_i^y \sigma_{i+1}^x) + \alpha_3 \sum_i (\sigma_i^y \sigma_{i+1}^z + \sigma_i^z \sigma_{i+1}^y) \quad (11)$$

To further increase the fidelity, a second order ansatz with next-nearest-neighbor terms is probed:

$$\begin{aligned} \mathcal{A}_{\text{next}} = & \alpha_1 \sum_i \sigma_i^y + \alpha_2 \sum_i (\sigma_i^x \sigma_{i+1}^y + \sigma_i^y \sigma_{i+1}^x) + \alpha_3 \sum_i (\sigma_i^y \sigma_{i+1}^z + \sigma_i^z \sigma_{i+1}^y) + \\ & + \alpha_4 \sum_i (\sigma_i^x \sigma_{i+2}^y + \sigma_i^y \sigma_{i+2}^x) + \alpha_5 \sum_i (\sigma_i^y \sigma_{i+2}^z + \sigma_i^z \sigma_{i+2}^y) \end{aligned} \quad (12)$$

To test the improvement of COLD in the context of QA, the commonly used metric is the fidelity of the prepared state  $\mathcal{F}_{COLD}$  after the optimization of the control parameter  $\beta$ . However, this metric itself is not sufficient to characterize the effectiveness of COLD, nor to justify its unwarranted use. For this reason, some additional quantities are going to be reported. The first one is the fidelity of the prepared state without QOC, thus using only LCD, which is  $\mathcal{F}_{LCD}$ . The effectiveness of the QOC optimization can be seen by comparing  $\mathcal{F}_{COLD}$  to  $\mathcal{F}_{LCD}$ . Another useful indicator is the preparation fidelity without any optimized schedule  $\mathcal{F}_{\text{std}}$ , i.e. the fidelity obtained driving the evolution using the bare Hamiltonian  $H_0(t)$  itself.

Having designated all the quantities above, a new success metric,  $\mathcal{R}$ , is defined as the ratio of fidelities between COLD and standard annealing schedules:

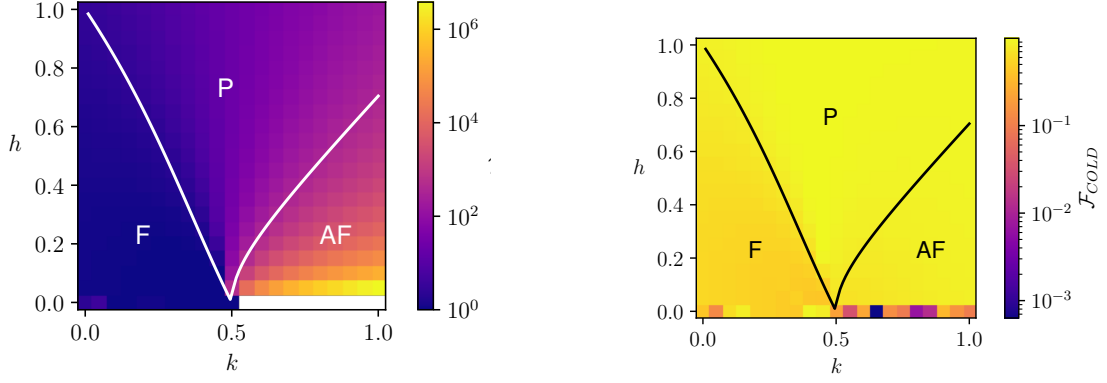
$$\mathcal{R} \equiv \mathcal{F}_{COLD} / \mathcal{F}_{\text{std}} . \quad (13)$$

An high  $\mathcal{R}$  is indicative of an higher relative improvement of fidelity with respect to trivial unoptimized schedules. This indicator does not replace the absolute fidelities  $\mathcal{F}_x$ , but is complementary. Indeed, a small relative improvement can still be a good result if the absolute fidelities are sufficiently high. Nevertheless, a bigger  $\mathcal{R}$  does not necessarily imply that COLD is effective, as it might happen that LCD alone is sufficient to achieve good fidelities, thus one has to consider the absolute fidelity of LCD before jumping to such conclusions.

### i. ANNNI model annealing

The ANNNI model is a variant of the Ising model which introduces next-nearest-neighbor couplings between spins. As previously stated, the target of our experiments is to prepare ground-states of the ANNNI model in the plane identified by the parameters  $h$  and  $k$  in the interval  $[0, 1]$ , fixing the coupling strength of the nearest neighbor spins to  $J^* = 1$ . Periodic conditions are assumed at





(a) Relative improvement of fidelity using COLD. Points for  $k > 0.5$  and  $h = 0$  are intentionally left blank as the relative improvement  $\mathcal{R}$  is greater than  $10^{30}$ .

(b) Absolute fidelity achieved by COLD.

**Figure 3:** Results of ground-state preparation of an ANNNI model in a QA setup. The Hamiltonian of Eq. 14 is prepared for different values of  $h$  and  $k$ . Simulations for  $N = 5$  spins, using the ansatz  $\mathcal{A}_{\text{next}}$ .

boundaries. Changing the target values of  $h$  and  $k$ , one could prepare three different phases of the ground-state: ferromagnetic (F), paramagnetic (P) and anti-ferromagnetic (AF).

The target Hamiltonian is written as

$$H_f^{(\text{ANNNI})} = -J^* \sum_i \sigma_i^x \sigma_{i+1}^x + k \sum_i \sigma_i^x \sigma_{i+2}^x + h \sum_i \sigma_i^z. \quad (14)$$

The set of operators that are actively controlled is given by a local field homogeneously applied to the spin chain.

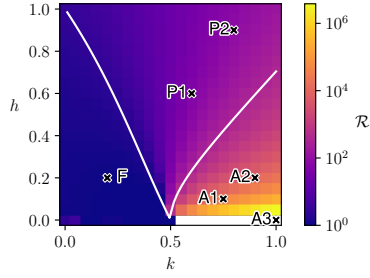
$$\mathcal{O}_{\text{opt}}^{(\text{ANNNI})} = \sum_i \sigma_i^z$$

Figure 3a plots the success metric  $\mathcal{R}$  for the ground-state preparation of the ANNNI model through different combinations of parameters. Remarkably, the greatest improvement is obtained in the preparation of the paramagnetic and anti-ferromagnetic phases, improving by 3 to 6 orders of magnitude. By a quick look at Figure 3b, it emerges that the absolute fidelity obtained using COLD  $\mathcal{F}_{\text{COLD}}$  is roughly homogeneous across all the phases of this model, with values greater than  $5 \cdot 10^{-1}$ .

In deeper analysis, some simulations on the phase diagram of Figure 3a have been selected. The sampled points metrics are reported in Table 1. The data hints that the ground-state in the ferromagnetic phase is not hard to be prepared, as the unoptimized schedule has already a fidelity of 0.5 that does not increment significantly by using either LCD or COLD.

Instead, a noticeable difference emerges in the paramagnetic phase, where unoptimized schedules present small absolute fidelities, which are slightly improved by LCD and are eventually maximized to  $\sim 0.95$  by COLD. As of the anti-ferromagnetic phase, the unoptimized annealing schedules exhibit almost null fidelities, which are boosted by COLD to  $\mathcal{F}_{\text{COLD}} \simeq 0.80$ . An extraordinary exception is the parameter combination A3, which lies in the region  $k > 0.5$  and  $h = 0$ . Those points are characterized by a  $\mathcal{F}_{\text{std}} \sim 10^{-33}$ , which is boosted to  $\mathcal{F}_{\text{COLD}} \sim 10^{-2}$ . Since the fidelity of LCD is equal to the bare annealing fidelity, we conclude that QOC is crucial to prepare anti-ferromagnetic ground-states in which the final transverse field is null ( $h = 0$ ).

In second analysis, it is necessary to test how the improvement scales with the system size  $N$ . Table 2 reports the success metric  $\mathcal{R}$  for  $N$  up to 11. Except for the A3 configuration, the improvement is



| label | $k$  | $h$ | $\mathcal{F}_{std}$     | $\mathcal{F}_{LCD}$     | $\mathcal{F}_{COLD}$ |
|-------|------|-----|-------------------------|-------------------------|----------------------|
| F     | 0.20 | 0.2 | 0.4897                  | 0.5001                  | 0.5230               |
| P1    | 0.60 | 0.6 | $1.7671 \cdot 10^{-2}$  | 0.2701                  | 0.9386               |
| P2    | 0.80 | 0.9 | $1.0881 \cdot 10^{-2}$  | 0.5031                  | 0.9609               |
| A1    | 0.75 | 0.1 | $2.0579 \cdot 10^{-5}$  | $3.8196 \cdot 10^{-4}$  | 0.7778               |
| A2    | 0.90 | 0.2 | $7.9326 \cdot 10^{-5}$  | $4.2983 \cdot 10^{-3}$  | 0.8525               |
| A3    | 1.00 | 0.0 | $3.0815 \cdot 10^{-33}$ | $3.0815 \cdot 10^{-33}$ | 0.0942               |

**Table 1:** ANNNI model, ground-state preparation for  $N = 5$  spins. Several simulations  $(k, h)$  are sampled from the phase diagram on the left, and the absolute fidelities are reported in the table.

| label | $\mathcal{R}$         |                       |                       |                       |                       |
|-------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|       | 4                     | 5                     | 7                     | 9                     | 11                    |
| F     | 1.437                 | 1.067                 | 1.000                 | 1.037                 | 1.000                 |
| P1    | 2.647                 | 53.12                 | $1.611 \cdot 10^2$    | $6.979 \cdot 10^2$    | $3.646 \cdot 10^3$    |
| P2    | 1.089                 | $3.780 \cdot 10^4$    | $8.700 \cdot 10^4$    | $6.319 \cdot 10^8$    | $2.119 \cdot 10^9$    |
| A1    | 6.350                 | 88.30                 | $2.827 \cdot 10^2$    | $2.124 \cdot 10^3$    | $1.044 \cdot 10^4$    |
| A2    | 1.510                 | $1.075 \cdot 10^4$    | $2.485 \cdot 10^4$    | $4.94 \cdot 10^7$     | $1.619 \cdot 10^8$    |
| A3    | $5.483 \cdot 10^{31}$ | $3.058 \cdot 10^{31}$ | $1.137 \cdot 10^{31}$ | $3.309 \cdot 10^{30}$ | $1.358 \cdot 10^{30}$ |

**Table 2:** ANNNI model,  $\mathcal{F}_{COLD}/\mathcal{F}_{std}$  for various  $N$ .

generally increasing with the system size. This is due to absolute fidelities of the unoptimized schedules that further decrease in larger system. The absolute COLD fidelity, instead, is roughly constant.

A final result is worth to be mentioned. In general, one could think that the COLD optimizations over the same system (and parameter settings) have to be carried out independently for different values of  $N$ . However, a pattern in the optimized parameters hints some form of *inheritance* of the optimized parameters towards systems of larger size.

This observation would be of critical importance in the optimization of larger systems, since the cost of each QA simulation can be computationally expensive. Indeed, it would be possible to inherit the  $\beta$  parameters from optimizations carried out on small values of  $N$ , eventually using such parameter on larger systems without executing the optimization routine from scratch.

This pattern has proved to be beneficial in the case of the ANNNI model. However, it is not guaranteed that it can be applied in any circumstance. Indeed, further tests on XXZ and Haldane-Shastry spin systems have proved that this inheritance mechanism is not always as effective as in the ANNNI model. Our conjecture is that the inheritance applicability depends on the quality of the landscape of the loss function and other spin systems and control operator choices might get more complicated to optimize.

To provide a successful example of inheritance, Table 3 shows the optimized parameter for the ANNNI model in the A1 configuration. The fidelities obtained with independent optimizations are just a small improvement over the inherited fidelities. Furthermore, the independently optimized parameter is close enough to the inherited parameter, hinting that the independent optimizations are a second order correction to the inherited parameter.

As a final test, we would like to give some heuristics on the ansatz choice. The following Figures, 4 and 5, show how fidelities change across two choices of the ansatz,  $\mathcal{A}_{local}$  and  $\mathcal{A}_{next}$ , respectively. To complete the picture, we show on the left side of the figures a new quantity,  $\mathcal{G}$ , which is defined on the schedule  $x$  as

$$\mathcal{G}_x = 1 - \varepsilon_x \quad (15)$$

where  $\varepsilon_x$  is the energy of the prepared state  $|\psi(\tau)\rangle$  normalized with respect to the energy spectrum of





| $N$ | inherited from $N = 5$ |                      | independent optimization |                      |
|-----|------------------------|----------------------|--------------------------|----------------------|
|     | $\beta$                | $\mathcal{F}_{COLD}$ | $\beta$                  | $\mathcal{F}_{COLD}$ |
| 6   | 3.8607                 | 0.71919              | 3.3080                   | 0.71932              |
| 7   |                        | 0.64658              | 3.1832                   | 0.64675              |
| 9   |                        | 0.26225              | 4.2011                   | 0.26228              |
| 11  |                        | 0.33197              | 3.5228                   | 0.33201              |

**Table 3:** Test of optimized parameter inheritance on the A1 configuration of the ANNNI model.

the final Hamiltonian:

$$\varepsilon_x \equiv \frac{\langle \psi(\tau) | H_f | \psi(\tau) \rangle_x - E_{min}}{E_{max} - E_{min}} . \quad (16)$$

Heuristically, a value of  $\mathcal{G}$  closer to 1 corresponds to a prepared state energy closer to the ground-state of the problem hamiltonian.

The fidelity is only a partial description of the prepared state quality, which inevitably degrades for larger spin systems. Instead, the information on the prepared state energy is complementary, and can be considered as well for the validation of the CD protocol.

The data shows that the prepared state is close to the ground state energy, with a value of  $\mathcal{G}_{COLD} \simeq 0.85$  if we use a simple local ansatz  $\mathcal{A}_{local}$  (Figure 4). Another ansatz choice, such as  $\mathcal{A}_{next}$  (Figure 5) boosts the agreement to the ground state energy up to  $\mathcal{G}_{COLD} \simeq 0.97$ . As a reference, unassisted protocols show values of this metric in the order of  $\mathcal{G}_{UA} \simeq 10^{-5}$ .

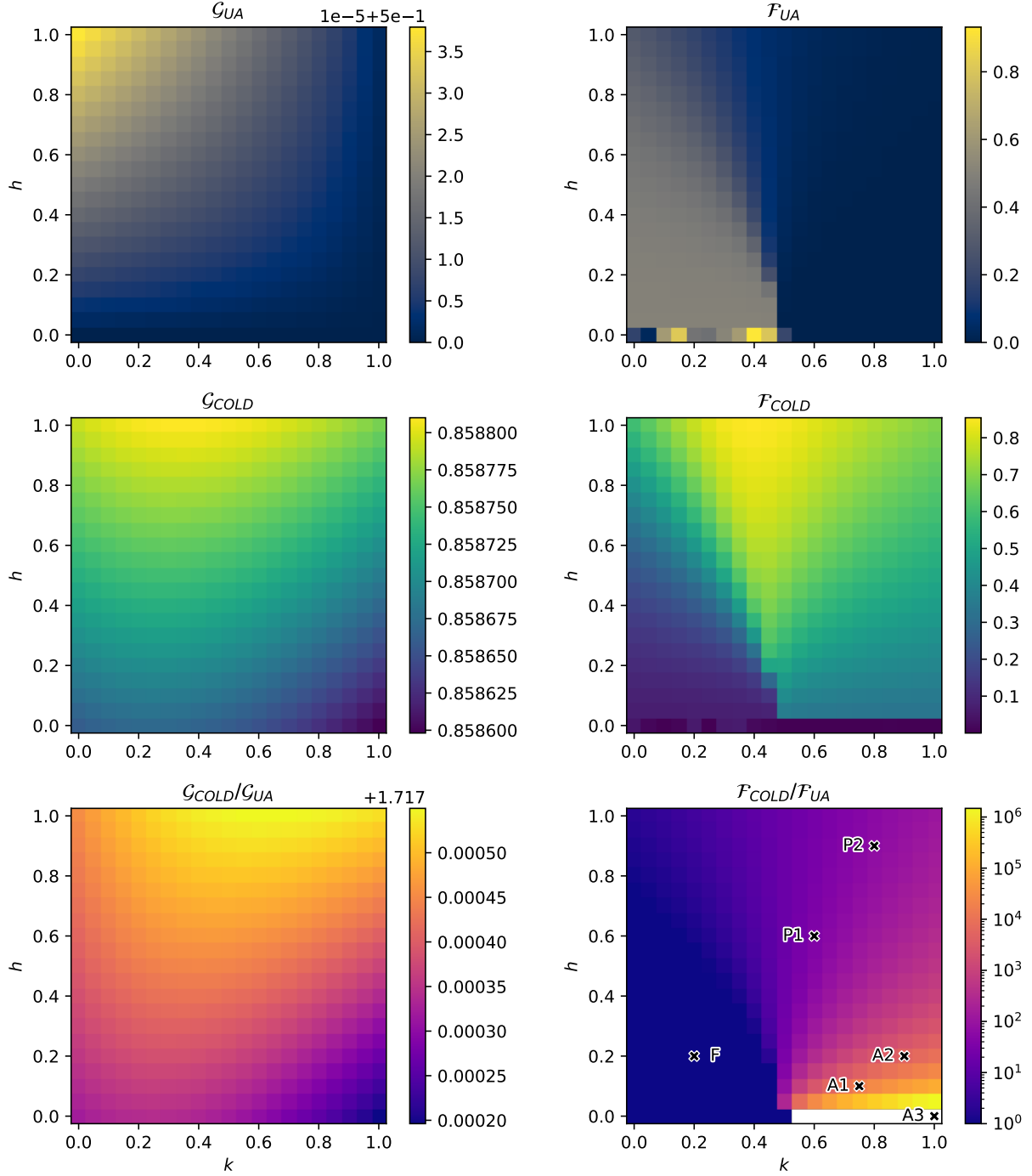
## 5 CONCLUSIONS

In this contribution, we focus on the optimization and benchmarking of counterdiabatic driving protocols (COLD) in models that exhibit phase transitions. Our findings indicate that COLD consistently outperforms standard annealing methods, often by orders of magnitude, and achieves nearly perfect fidelity when applied to antiferromagnetic states. Notably, COLD demonstrates good efficacy in addressing the ANNNI model, characterized by its non-integrable nature and diverse phase diagram. Furthermore, we show that optimized parameters from smaller models can be extended to higher-dimensional systems, emphasizing the pivotal role of optimized paths and adiabatic compensation in practical adiabatic state preparation applications. We introduce several enhancements, including the utilization of Bayesian optimization and higher-order approximate adiabatic gauge potentials, supported by a dedicated code package [3] for convenient evaluation. This tool can be useful in validating choices of the local AGP, which is usually motivated by hardware-based considerations.

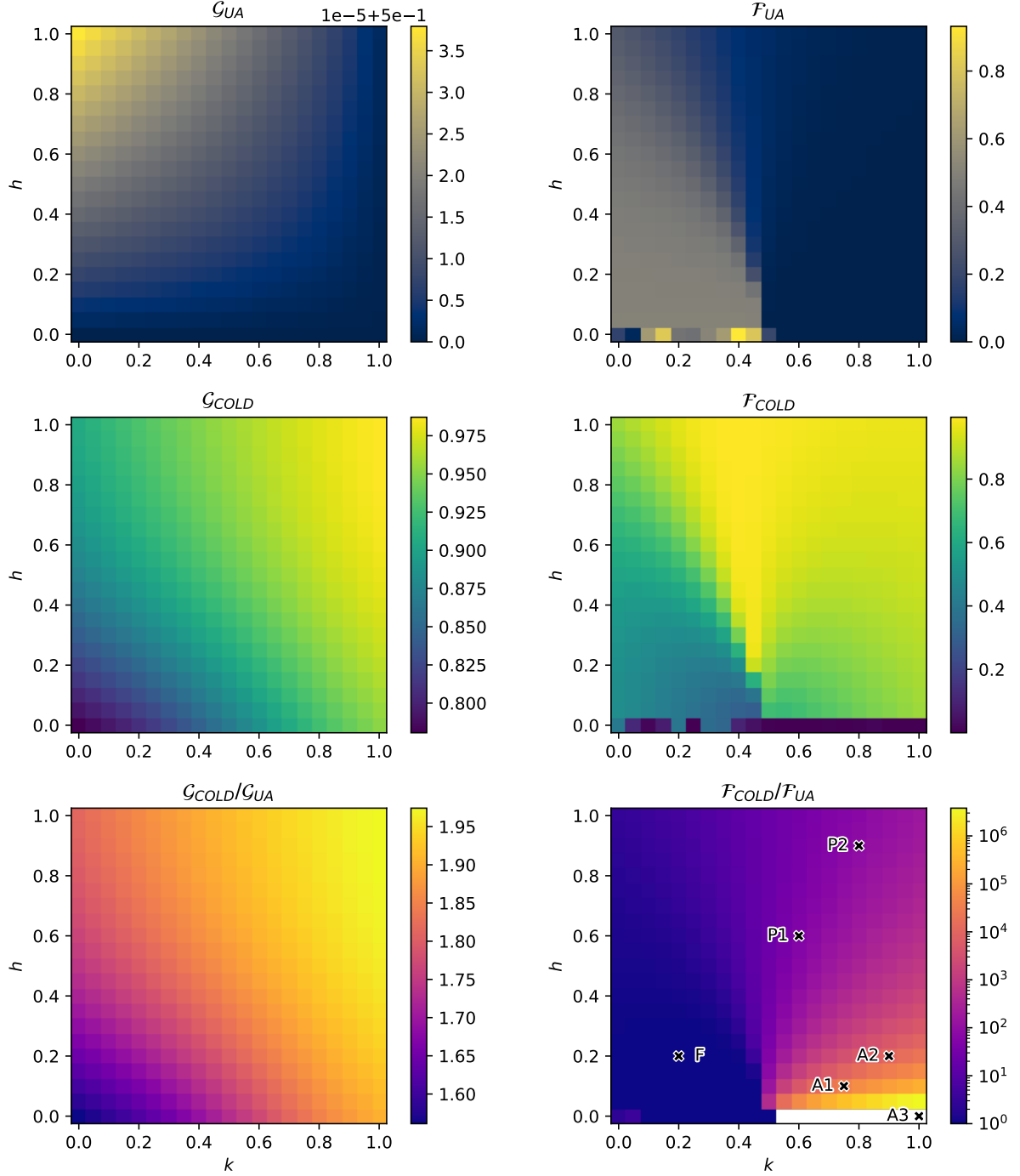
Counterdiabatic protocols have shown to be efficient solutions for optimal time-evolution of quantum systems. Furthermore, the combination with Optimal Control approaches in this task has been shown to improve the target fidelity beyond the values achieved by standard local counterdiabaticity. Nevertheless, the effectiveness depends on the specific spin system that is taken into consideration. Consequently, we advocate further research efforts aimed at identifying effective adiabatic gauge potentials for various models.







**Figure 4:** Various metrics for the state preparation of ANNNI model, using the local ansatz  $\mathcal{A}_{\text{local}}$ .



**Figure 5:** Various metrics for the state preparation of ANNNI model, using the next-nearest-neighbor ansatz  $\mathcal{A}_{\text{next}}$ .